

AE-591: Stocksolution of [Ru(CO)(OH)2(PNP{tBu})] in water

Date: 2025-10-14

Tags: PNP Degassed Stocksolution
P(tBu)N(py)P(tBu) [Ru(CO)(OH)2(PNP)]
AE

Category: Two-photon

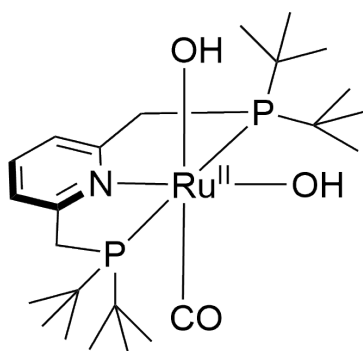
Status: Done

Created by: Alexander Eith

Objective

Making concetrated stock solution of [Ru(CO)(OH)2(PNP)] in water

Reaction scheme/sample structure



Chemical Formula: C₂₄H₄₅NO₃P₂Ru

Molecular Weight: 558,64552

ChemDraw file: [AE-493.cdxml](#)

Literature/reference experiments

Literature	/
Reproduction	/
Related experiment	Two-photon - AE-521: Stocksolution of [Ru(CO)(OH)2(PNP{tBu})] in water I Two-photon - AE-581: Stocksolution of [Ru(CO)(OH)2(PNP{tBu})] in water

Reagents

Name	CAS/Experiment number	Amount [μmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume [ml]	Density [g/mL]
[Ru(CO)(OH)2(PNP{tBu})]	Two-photon - AE-570: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, small batch	9.6	/	/	5.37	558.65	/	/

Milli-Q water	Prep work - AE-549: Degassing of Milli-Q water	/	/	/	/	/	5.4	0.998
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Procedure/observations

All steps were done according to [\[Protocol\] Schlenk Technique](#).

All steps were carried out in the dark.

Date	Time	Step	Observations	Pictures
13.10		A 10 mL Schlenk flask was set under argon.	/	/
	15:35	[Ru] was added in air	/	/
		The flask was set under argon	/	/
	15:55	To the flask water was added	yellow-orange solution	/
		The mixture was stirred at 500 rpm	no change	Before filt.jpg
		A 25mL schlenk flask was prepared	/	/
	16:12	The mixture was transferred into the new schlenk flask and thereby filtered thorough a PA syringe filter (pore diamter = 0.2 µm)	significatnly lighter in tone, hue did not change	/
	16:20	The flask was stored at -20 °C in the dark under argon in Equipment - Freezer Lab E004 CEEC II	AE-591	/

Product characterization

Sample	Volume [mL]	Purity	Conecntration [mM]	Description	Photo	Storage location
AE-591	5.4	98 %	concentrated solution (aprox. 1.44)	light yellow solution		Equipment - Freezer Lab E004 CEEC II

Result

Worked as expected

Linked experiments

Prep work - [AE-549: Degassing of Milli-Q water](#)

Two-photon - [AE-521: Stocksolution of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\] in water I](#)

Two-photon - [AE-570: Synthesis of \[Ru\(CO\)\(OH\)2\(PNPtBu\)\] using Ag2O, small batch](#)

Two-photon - [AE-581: Stocksolution of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\] in water](#)

Linked resources

Equipment - [Freezer Lab E004 CEEC II](#)

Protocol - [Schlenk Technique](#)

Attached file

Before filt.jpg

sha256: cfd6cf6ebbc79284c40c736ab85c96761c8183421ea975fb9afc73faff2bd310



Unique eLabID: 20251014-b809c36a0fb576f2c8de3552593f5bfe8224eae
Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=3183>